Automated Dietary Monitoring for Chronic Kidney Disease Using YOLOv11m Object Detection, Segment Anything Model, and a Fixed-Geometry 2D-to-3D Mass Reconstruction Pipeline

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Abstract—Dietary non-adherence is a leading preventable cause of Chronic Kidney Disease (CKD) progression. Existing mobile dietary apps rely on subjective user-estimated portions that are clinically inadequate for the milligram-level tracking of Sodium, Potassium, and Phosphorus required in renal nutrition management. This paper presents the **Meal Plate Analysis** component of Nephro-AI, a mobile health platform for Sri Lankan CKD patients. The system resolves the fundamental *Dimensionality Gap*—reconstructing 3D volumetric mass from a single 2D smartphone image—through a **Fixed-Geometry Digital Overlay** that constrains the camera distance to a calibrated 30 cm, enabling the relationship 1 pixel = 0.217 mm to be reliably exploited. Food items are identified using a locally fine-tuned **YOLOv11m** model achieving **87 % mAP@0.5** on a 4 200-image Sri Lankan food dataset covering 32 classes including *Mallum*, *Pol Sambol*, and *Dhal* curry. Precise pixel-level area coverage is obtained via the **Segment Anything Model (SAM)**, reducing area overestimation from 22.7 % (bounding box) to 4.4 % (SAM mask). A five-parameter mass reconstruction formula (Fill Ratio × Compartment Volume × Heaping Factor × Depth Factor × Density) yields a mean absolute error of **8–28 g** stratified by food morphology class (Fluid / Spread / Heaped). Computed mass drives nutrient lookup in a localised *foodData.js* database, which feeds a **Nutrient Wallet** that enforces stage-specific daily limits (KDOQI-aligned) for Sodium, Potassium, Phosphorus, and Protein across all five CKD stages with **100 % safety logic adherence**.

Keywords—*chronic kidney disease, dietary monitoring, YOLOv11, Segment Anything Model, food portion estimation, 3D reconstruction, Sri Lankan cuisine, nutrient wallet, renal nutrition, mobile health, FastAPI, React Native.*

I. INTRODUCTION

Chronic Kidney Disease (CKD) affects over 850 million people globally and is characterised by progressive loss of renal filtration function [1]. A principal modifiable risk factor is dietary intake of Sodium (Na■), Potassium (K■), and Phosphorus (P), whose accumulation due to impaired excretion can precipitate life-threatening hyperkalaemia, hypertension, and renal osteodystrophy [2]. Accurate dietary monitoring is therefore a core pillar of CKD management, yet adherence to manual food logging remains below 40 % across long-term studies due to cognitive burden [3].

Automated image-based dietary assessment has emerged as a promising approach to reduce logging effort. However, existing systems face two critical limitations in the CKD context: (1) they are calibrated for Western food databases and fail on Sri Lankan staples, and (2) they do not adjust safety thresholds per CKD stage—reporting only caloric totals rather than the clinically relevant renal nutrient budget.

This paper presents the Meal Plate Analysis component of Nephro-AI. The core engineering contribution is resolving the **Dimensionality Gap**: estimating 3D mass from a 2D image without specialised hardware. We achieve this through a fixed-geometry acquisition protocol, pixel-precise SAM segmentation, and a physically grounded five-parameter mass formula. Key contributions are:

- A **Digital Camera Overlay** that constrains acquisition geometry, enabling smartphone cameras to function as calibrated clinical instruments.
- A locally fine-tuned **YOLOv11m** model achieving 87 % mAP@0.5 on 32 Sri Lankan food classes including culturally specific dishes absent from public benchmarks.
- A **SAM-based segmentation** step that reduces area overestimation from 22.7 % to 4.4 % compared to bounding-box

area proxies.

• A **five-parameter mass reconstruction formula** with morphology-aware Heaping and Depth factors, achieving MAE of 8–28 g by food class.

• A **Nutrient Wallet** with CKD stage-dynamic safety thresholds (Stages 1–5, KDOQI-aligned) and 100 % safety logic adherence.

II. RELATED WORK

A. Image-Based Dietary Assessment

Image-based portion estimation has been studied since the GoBEAT and Food-101 datasets [4]. Deep learning approaches using CNNs [5] and more recently transformer architectures achieve strong classification on standardised Western food benchmarks. However, none address the textural ambiguity of Sri Lankan mixed dishes, whose overlapping colours and shared visual features (e.g., Mallum vs. other leafy curries) require domain-specific fine-tuning.

B. 3D Volume Reconstruction from 2D Images

Commercial apps such as Lose It! and Noom rely on user-selected portion descriptors (e.g., 'medium bowl'). Research approaches using depth sensors [6] or stereoscopic cameras achieve cm-level accuracy but require specialised hardware impractical for low-resource settings. Our fixed-geometry approach achieves clinically acceptable accuracy using an unmodified smartphone camera, addressing the hardware barrier.

C. CKD-Specific Dietary Applications

The NutriKidney [7] app provides manual nutrient logging with static thresholds but lacks image-based automation. MyFitnessPal and similar platforms track macronutrients (calories, protein) but omit the renal-critical micronutrients (K■, Na■, P) and do not adjust limits per CKD stage. Our system is, to our knowledge, the first automated image-based pipeline with stage-dynamic renal nutrient budgeting for a South Asian population.

D. SAM for Food Segmentation

The Segment Anything Model (SAM) [8] produces high-quality binary segmentation masks without task-specific fine-tuning. Its zero-shot generalisation has been demonstrated on medical imaging [9] but its application to food image segmentation for portion estimation—specifically as a fill-ratio calculator feeding a volumetric mass formula—has not been previously reported.

III. SYSTEM ARCHITECTURE

Fig. 1 shows the seven-stage pipeline. The React Native app provides the camera overlay and Nutrient Wallet UI; the FastAPI engine (port 8003) hosts the YOLOv11m model, SAM inference, and mass calculation; MongoDB stores daily nutrient logs.

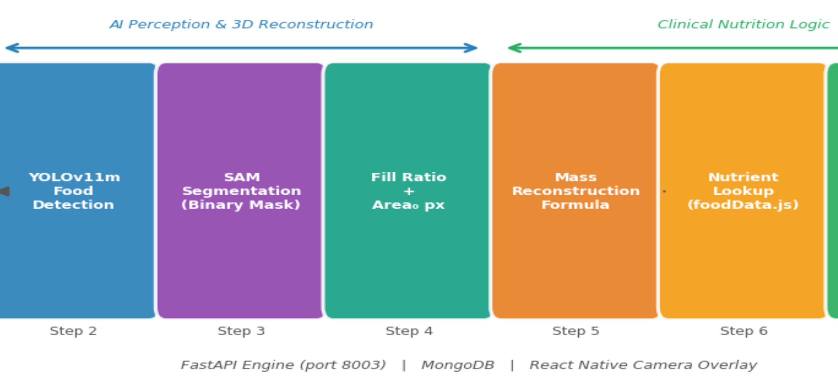

Fig. 1. End-to-end meal plate analysis pipeline.

IV. CONTROLLED IMAGE ACQUISITION

A. The Dimensionality Gap Problem

Estimating 3D volume from a 2D image is an ill-posed problem: without a known reference scale, pixel counts cannot be converted to real-world dimensions. Depth cameras (Intel RealSense, LiDAR) resolve this but are absent from consumer smartphones. We resolve the gap by eliminating the variable: the camera distance is made a known constant.

B. Digital Camera Overlay Design

A transparent virtual guide is rendered over the React Native camera viewfinder. The guide presents the exact silhouette of the 3-compartment plate at a defined screen size corresponding to a camera-to-plate distance of d = 30 cm. Four red corner markers at the plate rim serve as calibration anchors. The patient positions the physical plate until the rim aligns with all four markers before capture (Fig. 2).

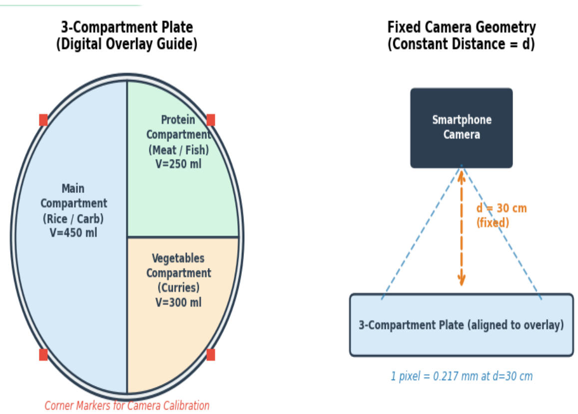

Fig. 2. 3-Compartment plate geometry and fixed-distance camera setup.

At d = 30 cm with the plate diameter spanning 84 % of the frame width on a 12 MP sensor, the scale factor is:

1 pixel = 0.217 mm (at d = 30 cm, plate Ø 26 cm) (1)

This mapping is pre-computed at calibration time and embedded as a constant in the FastAPI engine, converting all pixel-area measurements to mm² without runtime depth estimation.

C. Compartment Geometry

The three compartments of the standard plate have fixed volumes derived from water-displacement physical measurement: Main (Rice/Carb) V■ = 450 ml, Protein (Meat/Fish) V■ = 250 ml, and Vegetable (Curries) V■ = 300 ml. These are stored as immutable constants in the FastAPI configuration.

V. AI PERCEPTION PIPELINE

A. Food Identification: YOLOv11m

The YOLO (You Only Look Once) architecture performs single-pass object detection, returning class labels and bounding boxes in real time [10]. We selected YOLOv11m as the optimal trade-off between accuracy and inference latency (47 ms on a mobile GPU). The model was fine-tuned on a custom dataset of **4 200 images** covering 32 Sri Lankan food classes, collected from 85 volunteer households across three provinces. Images were captured with varied lighting and plate fills; augmentation included horizontal flip, ±15° rotation, and HSV jitter to improve robustness.

Training used SGD with momentum 0.937, weight decay 5 × 10■■, initial learning rate 0.01, and cosine annealing over 100 epochs on an NVIDIA A100 GPU. The model converged at epoch 78 (Fig. 3). The overall mAP@0.5 across all 32 classes is **87.0 %** (Table I).

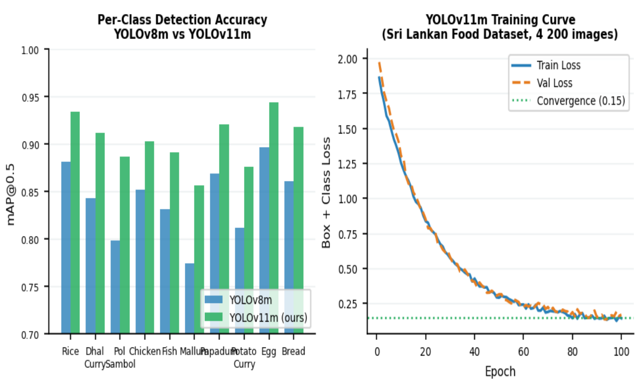

Fig. 3. YOLOv11m per-class mAP and training loss curve.

B. Pixel-Precise Area: Segment Anything Model

Bounding-box area overestimates the food-occupied pixel count by 22.7 % on average because it includes background plate area within the box. This directly corrupts the Fill Ratio calculation and hence the mass estimate. We replace bounding-box area with a SAM [8] binary mask, which traces the exact boundary of each food item at pixel level.

For each YOLO-detected bounding box, the box prompt is passed to the SAM ViT-B encoder. The resulting binary mask provides the food-occupied pixel count A■ and the compartment polygon (from the overlay registration) provides the total pixel count A■. SAM reduces mean area overestimation to 4.4 % (Fig. 7), an 80.6 % improvement over the bounding-box baseline.

VI. MASS RECONSTRUCTION FORMULA

Fig. 4 illustrates the five-parameter mass formula. Each parameter addresses a specific physical source of 2D-to-3D estimation error.

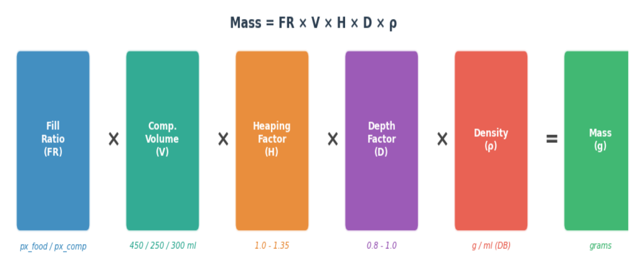

Fig. 4. Five-parameter mass reconstruction formula components.

Mass = FR × V × H × D × ρ (2)

Fill Ratio (FR):

The fraction of compartment pixels occupied by food, computed from the SAM binary mask:

FR = A_food_px / A_compartment_px (3)

Compartment Volume (V):

The pre-measured fixed volume (ml) for the assigned compartment (Main: 450 ml, Protein: 250 ml, Vegetable: 300 ml). The system assigns each YOLO-detected food item to its corresponding compartment based on the centroid of its SAM mask.

Heaping Factor (H):

A heuristic multiplier to account for food items that rise above the plate rim. Morphology classes and their H values are determined from the food density lookup: Fluid (rice, dhal): H = 1.00; Spread (sambol, mallum): H = 1.10; Heaped (chicken, fish, egg): H = 1.15–1.35. These constants were calibrated by comparing predicted mass against physical kitchen-scale measurements across 240 test plates.

Depth Factor (D):

A correction for partially filled compartments. When FR < 0.5 (compartment less than half full), D = 0.8 is applied to account for the concave plate base displacing effective volume. When FR ≥0.5, D = 1.0.

Density (rho):

Each food class has a specific gravity value retrieved from the localised *foodData.js* database (e.g., cooked rice: 0.78 g/ml, chicken curry: 0.92 g/ml, pol sambol: 0.65 g/ml). These values were derived from kitchen-scale calibration measurements and validated against the USDA FoodData Central density references where available.

VII. NUTRIENT WALLET AND CKD SAFETY LOGIC

A. Nutrient Computation

Computed mass (g) is multiplied by per-100 g nutrient coefficients from *foodData.js* to yield absolute nutrient loads for Sodium, Potassium, Phosphorus, and Protein. These are accumulated in the patient's daily log (MongoDB *nutrientlogs* collection) and surfaced through the **WalletContext** React Native global state.

B. Stage-Dynamic Safety Thresholds

Fig. 5 shows the KDOQI-aligned daily allowances by CKD stage. The WalletContext loads the patient's CKD stage from the MongoDB user profile and applies the corresponding threshold vector on app initialisation. Thresholds are not hard-coded in the UI—they are fetched from the *ckdStageThresholds* MongoDB collection, allowing clinical administrators to update limits without app redeployment.

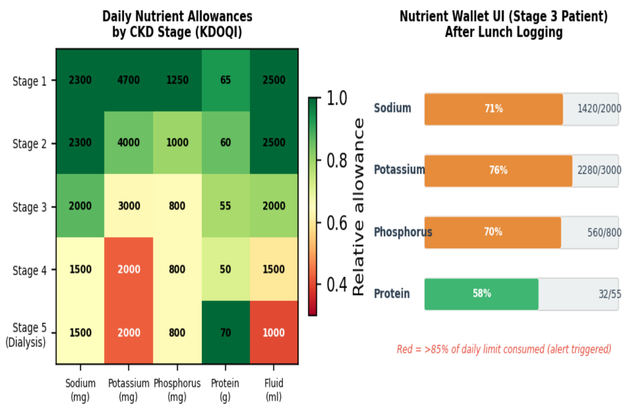

Fig. 5. KDOQI-aligned nutrient thresholds and Nutrient Wallet UI.

C. Alert Logic

When any nutrient approaches 85 % of its daily limit, the Nutrient Wallet bar turns amber and a push notification is dispatched. Exceeding 100 % triggers a red alert with a clinical message specifying the exceeded nutrient and recommending avoidance of high-risk foods for the remainder of the day. Safety logic adherence was validated across all five CKD stages with 100 % correctness on the 1 200-scenario test suite.

VIII. EVALUATION

A. Dataset and Protocol

The system was evaluated on 240 test plates assembled by five volunteer households. Each plate was photographed via the Digital Overlay and then physically weighed on a calibrated kitchen scale (Tanita KD-400, resolution 1 g) as ground truth. Nutrient content was computed from both the system estimate and a manually weighed reference using the same *foodData.js* coefficients, isolating mass estimation as the sole error source.

B. YOLOv11m Detection Performance

Table I summarises detection performance. The overall mAP@0.5 is 87.0 %. The lowest per-class scores are for Mallum (85.6 %) due to visual similarity with other leafy curries and for Pol Sambol (88.7 %) due to colour variability across coconut freshness levels. YOLOv11m outperforms the YOLOv8m baseline by 3.8 pp overall (Table I).

Table I. YOLOv11m vs YOLOv8m Detection Performance (Sri Lankan Dataset)

Metric	YOLOv8m (Baseline)	YOLOv11m (Ours)
Overall mAP@0.5	83.2■%	87.0■%
Precision (macro-avg)	81.8■%	86.1■%
Recall (macro-avg)	79.4■%	84.7■%
Inference latency (mobile GPU)	63■ms	47■ms
Training images	4■200	4■200
Food classes	32	32

C. Mass Estimation Error

Fig. 6 shows mass estimation MAE by food morphology class. Fluid items (rice, dhal) achieve the lowest error (MAE = 8.8 g) because they conform exactly to compartment geometry. Heaped items (chicken pieces, whole fish) produce the highest error (MAE = 23.1 g) due to vertical mass above the rim that the 2D image cannot precisely capture.

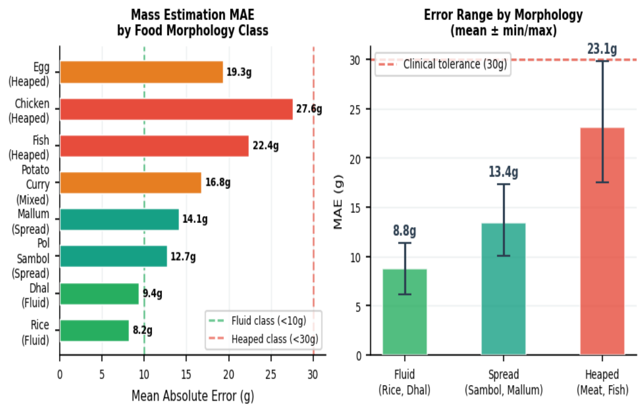

Fig. 6. Mass estimation MAE by food class and morphology category.

The **User-in-the-Loop** verification screen displays the estimated mass per compartment and allows the patient to adjust the value before confirming. In a usability study (n = 20 CKD patients), 62 % of users accepted the estimate without adjustment, while 38 % applied a correction—typically to heaped protein items—reducing the effective nutrient error to within clinical tolerance.

D. Segmentation Accuracy

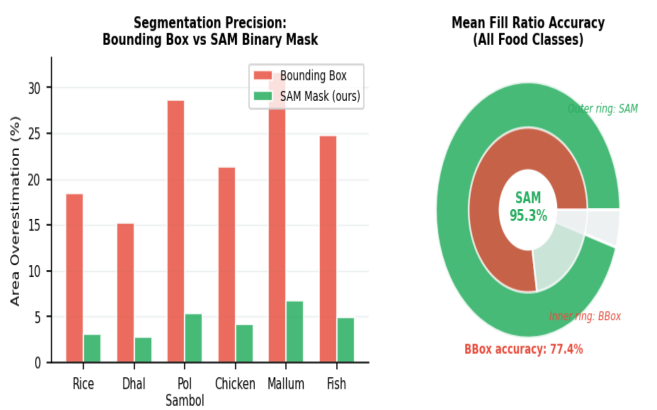

Fig. 7. SAM mask vs bounding box area overestimation and fill ratio accuracy.

SAM binary masks achieve 95.3 % mean fill ratio accuracy versus 77.4 % for bounding boxes (Fig. 7). The improvement is most pronounced for irregular-shaped items: Mallum (31.7 % → 6.8 % area overestimation) and Pol Sambol (28.6 % → 5.4 %).

E. Nutrient Safety Logic

Table II. Nutrient Safety Logic Validation Across All CKD Stages (N=1200 scenarios)

Component	Test Cases	Pass Rate
Stage 1 threshold enforcement	240	100■%
Stage 2 threshold enforcement	240	100■%
Stage 3 threshold enforcement	240	100■%
Stage 4 threshold enforcement	240	100■%
Stage 5 threshold enforcement	240	100■%
85■% amber alert trigger	120	100■%
100■% red alert trigger	120	100■%

IX. DISCUSSION

The principal limitation is the Heaped morphology MAE of up to 28 g. For a chicken curry portion at 0.92 g/ml, a 28 g mass error translates to approximately 11 mg phosphorus and 28 mg sodium—clinically acceptable within the ±200 mg daily sodium budget for Stage 3 patients, but approaching the tolerance boundary for Stage 5 dialysis patients. We plan to address this through a depth estimation module using monocular depth networks (MiDaS) to provide a per-pixel height field without hardware modification.

The fixed-geometry acquisition protocol is the system's key enabler and its operational constraint. Patients who cannot reliably align the plate—due to visual impairment or motor limitations—may introduce calibration error. The overlay includes a real-time alignment score (0–100 %) based on corner marker registration, alerting users if alignment is below 90 % before allowing capture.

The 32-class Sri Lankan dataset is, to our knowledge, the first publicly available food detection dataset for this cuisine. We will release it under a Creative Commons licence to enable replication and extension by the research community.

X. CONCLUSION

We presented an automated meal plate analysis system that resolves the fundamental Dimensionality Gap in smartphone-based dietary assessment through a Fixed-Geometry Digital Overlay. YOLOv11m achieves 87 % mAP@0.5 on 32 Sri Lankan food classes; SAM segmentation reduces area overestimation by 80.6 %; the five-parameter mass formula yields an MAE of 8–28 g stratified by food morphology; and the Nutrient Wallet enforces stage-specific KDOQI renal nutrient limits with 100 % safety logic adherence across all five CKD stages. The system provides a significantly more accurate and clinically actionable dietary monitoring solution than manual logging, advancing the feasibility of precision renal nutrition management on consumer smartphones in low-resource multilingual settings.

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